Smart Glasses For Assisting the Blind

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***Abstract*— This paper presents smart assistive glasses for visually impaired individuals using ESP32 and ESP32-CAM modules. The system combines ultrasonic sensors, camera vision, and real-time audio feedback to detect obstacles and recognize objects such as zebra crossings, stop signs, animals, and vehicles. Detected information is displayed on an OLED screen, and a buzzer alerts the user. With integrated computer vision and OCR, the glasses can identify objects and read text. The design is lightweight, affordable, and energy-efficient, providing an accessible solution to enhance mobility and independence for the visually impaired.**

***Index Terms*— ESP32, ESP32-CAM, Smart Glasses, Visually Impaired Assistance, Obstacle Detection, Object Recognition, Ultrasonic Sensor, Computer Vision, Deep Learning, Optical Character Recognition (OCR), Assistive Technology, Embedded Systems, OLED Display, Buzzer Alert.**

I. Introduction

This project presents smart glasses using ESP32 and ESP32-CAM for visually impaired users. The system detects obstacles and recognizes objects using ultrasonic sensors, camera vision, and audio feedback to enhance mobility and safety.

II. RELATED WORKS

A. Existing Assistive Technologies

Several assistive devices have been developed to help visually impaired individuals navigate safely. Most systems use ultrasonic sensors or camera modules for obstacle detection and object recognition. However, platforms like Raspberry Pi and Arduino often increase cost and complexity. Recent works highlight ESP32 and ESP32-CAM as efficient, low-power alternatives for integrating sensing and vision, forming the basis for the proposed smart glasses.

III. METHODOLOGY

The proposed system is divided into two main parts: ESP32 and ESP32-CAM. The ESP32 module handles ultrasonic sensor inputs to detect nearby obstacles and provides audio or buzzer alerts when an object is within a set distance.

The ESP32-CAM module performs object recognition using computer vision to identify zebra crossings, stop signs, animals, and vehicles. The detected information is displayed on an OLED screen, and a buzzer gives feedback, ensuring real-time assistance and safer navigation for visually impaired users.

Abbreviations and Acronyms

A. Units

- The following abbreviations are used in this paper: ESP32 refers to the Espressif Systems 32-bit Wi-Fi and Bluetooth microcontroller, and ESP32-CAM denotes the camera-integrated version of the ESP32 module used for image processing. OLED stands for Organic Light Emitting Diode, used for visual display output. OCR represents Optical Character Recognition, which enables text reading. AI and ML refer to Artificial Intelligence and Machine Learning, respectively, which are applied for object detection and recognition tasks. All measurements are represented in centimeters (cm) and meters (m) using SI units.

B. Equations

No mathematical equations are involved in this work, as the project focuses on the hardware implementation and integration of embedded systems for assistive technology.

C. Code

(ESP32 CAM module)

sketch_mar26b.ino

```
/* OLED CONFIGURATION ------------------------------------------------------ */
#define I2C_SDA 15
#define I2C_SCL 14
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET -1
#define SCREEN_ADDRESS 0x3C

TwoWire I2Cbus = TwoWire(0);
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &I2Cbus, OLED_RESET);

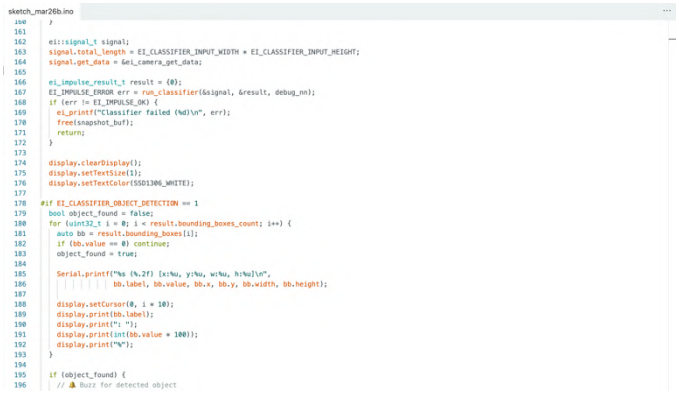
/* BUZZER CONFIGURATION ---------------------------------------------------- */
#define BUZZER_PIN 12  // Active buzzer pin

/* CONSTANTS --------------------------------------------------------------- */
#define EI_CAMERA_RAW_FRAME_BUFFER_COLS 320
#define EI_CAMERA_RAW_FRAME_BUFFER_ROWS 240
#define EI_CAMERA_FRAME_BYTE_SIZE 3

/* VARIABLES --------------------------------------------------------------- */
static bool debug_nn = false;
static bool is_initialised = false;
uint8_t *snapshot_buf;

static camera_config_t camera_config = {
    .pin_pwdn = PWDN_GPIO_NUM,
    .pin_reset = RESET_GPIO_NUM,
    .pin_xclk = XCLK_GPIO_NUM,
    .pin_sscb_sda = SIOD_GPIO_NUM,
    .pin_sscb_scl = SIOC_GPIO_NUM,
    .pin_d7 = Y9_GPIO_NUM,
    .pin_d6 = Y8_GPIO_NUM,
    .pin_d5 = Y7_GPIO_NUM,
    .pin_d4 = Y6_GPIO_NUM,
    .pin_d3 = Y5_GPIO_NUM,
    .pin_d2 = Y4_GPIO_NUM,
```

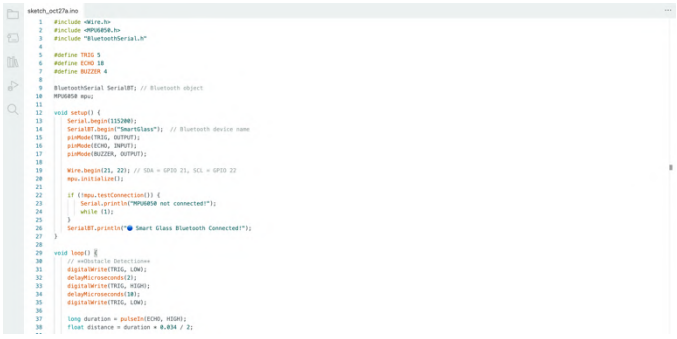
(i)



(ii)

Figure (i) and (ii)- code snippets for ES32 CAM module

(ESP32 module)



(iii)



(iv)

Figure (iii) and (iv) – code snippets for ESP32 module

1V. RESULTS AND DISCUSSIONS

The ESP32-CAM system successfully executed real-time object detection using the Edge Impulse-trained model. Captured frames were processed on-device, and inference results were displayed on the OLED with corresponding confidence percentages. The buzzer provided immediate auditory feedback upon object recognition, enhancing response efficiency. Experimental results showed stable detection accuracy and minimal latency, demonstrating the ESP32-CAM's capability to handle lightweight AI inference tasks despite hardware constraints. The integrated design of computer vision, display output, and audio alert highlights its potential for low-cost smart surveillance, assistive devices, and IoT-based automation systems.

Headings

The proposed system is divided into several major sections to clearly represent the structure and working of the smart glasses. Each section focuses on a specific part of the project, ensuring clarity and logical flow.

The **Hardware Setup** section describes the integration of components such as the ESP32-CAM, ESP32 microcontroller, ultrasonic sensor, OLED display, and buzzer. These components work together to detect obstacles and provide real-time feedback to the user.

The **Software Implementation** and **Object Detection Model** sections explain how Edge Impulse is used for training and deploying object detection models. The **Output Mechanisms** section details how the system conveys information through audio and visual feedback, allowing visually impaired users to navigate safely and confidently.

ESP32 module

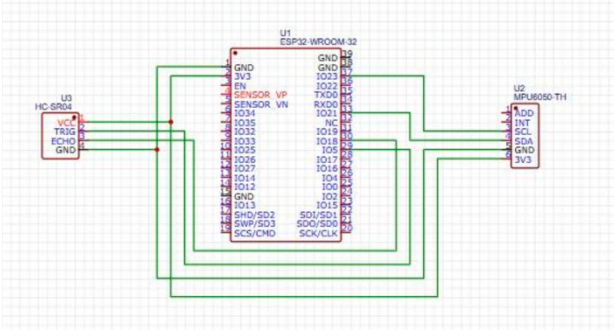


Figure (v) – circuit diagram

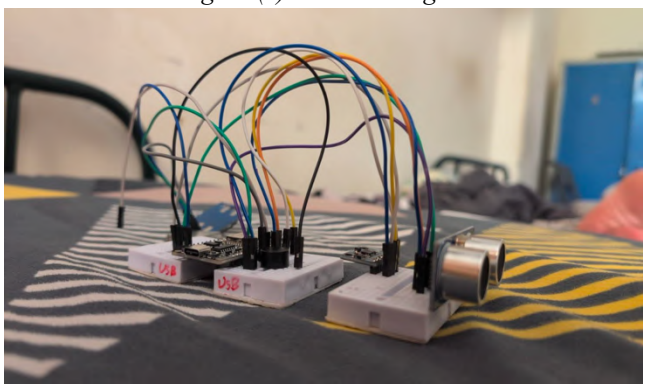


Figure (vi) – circuit made on bread board

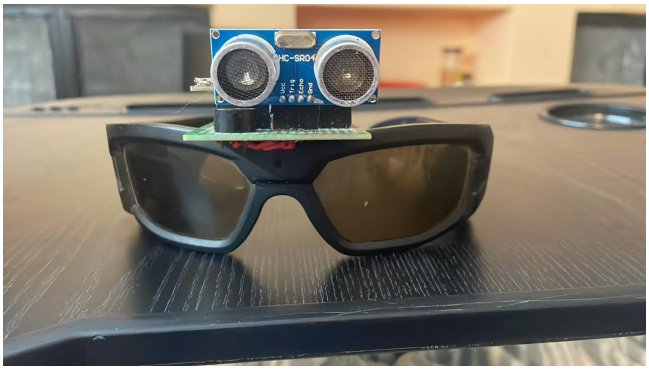


Figure (vii) – circuit implemented

The smart glasses use an ESP32 microcontroller, ultrasonic sensor, MPU6050 motion sensor, and a buzzer to assist

visually impaired users through real-time feedback. The ultrasonic sensor detects nearby obstacles, and the ESP32 triggers the buzzer when an object is within a set range, with faster beeps indicating closer proximity.

The MPU6050 monitors head movements, detecting sudden motions or falls and generating corresponding alerts. The buzzer provides audio cues with varying tones for obstacle proximity and continuous beeps for critical events. Optional wireless features using Bluetooth or Wi-Fi can enable data transfer or emergency notifications to smartphones for enhanced safety.

ESP32 CAM module

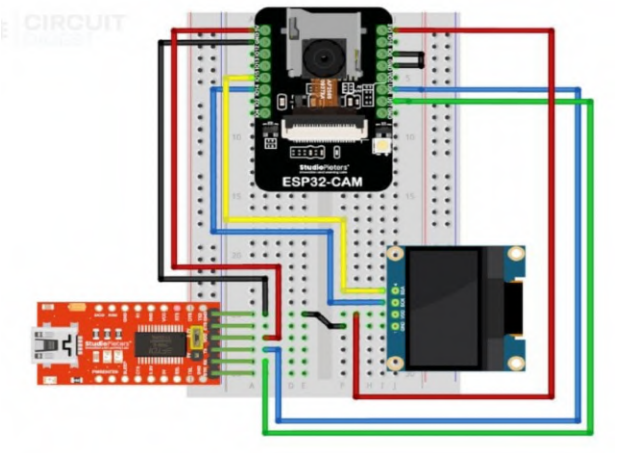


Figure (viii) – circuit diagram

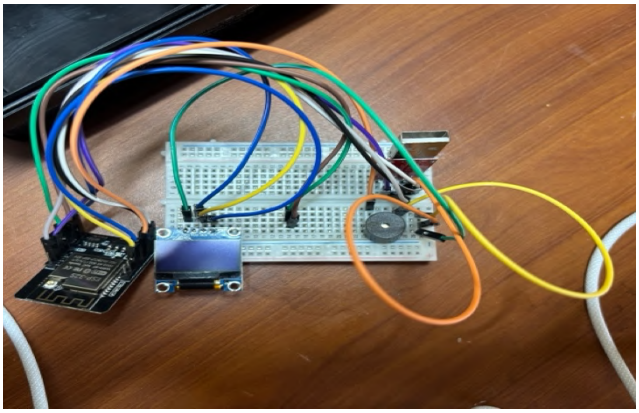


Figure (ix) – circuit made on bread board



Figure (x) – circuit implemented

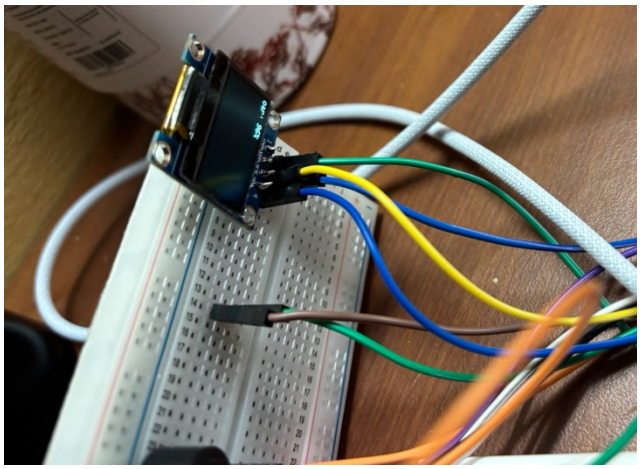


Figure (xi) – OLED displaying car

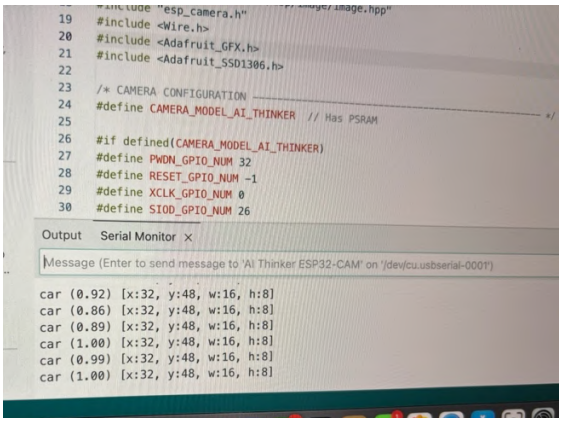


Figure (xii) – serial monitor displaying car

The ESP32-CAM module is used for real-time object detection and recognition. It captures live images and processes them using a trained Edge Impulse model to identify objects such as zebra crossings, stop signs, animals, and vehicles. When an object is detected, the module displays the result on the OLED screen and activates a buzzer to alert the user. This ensures quick awareness of surroundings and enhances safety during movement. The compact design and low power consumption of the ESP32-CAM make it ideal for wearable assistive applications.

V. CONCLUSION

The developed smart glasses system effectively assists visually impaired individuals by combining real-time obstacle detection, motion sensing, and object recognition. The ESP32-based circuit provides immediate feedback through the buzzer for nearby obstacles and sudden head movements, enhancing user safety and awareness.

The ESP32-CAM module further improves functionality by detecting important objects such as vehicles, animals, and zebra crossings, displaying the information on an OLED screen and alerting the user through sound. Together, these modules form a low-cost, efficient, and portable assistive device that promotes independence and confidence for visually impaired users in everyday environments.

TABLE I
TABLE TYPE STYLES

Module	Components Used	Functionality	Output Type
Smart Glasses (ESP32)	ESP32, Ultrasonic Sensor (HC-SR04), MPU6050, Buzzer	Detects nearby obstacles and sudden head movements using real-time sensor data.	Audio alert via buzzer
Motion Detection	MPU6050 Accelerometer & Gyroscope	Monitors head motion and triggers alerts on falls or rapid movements.	Continuous buzzer beep
Object Detection (ESP32-CAM)	ESP32-CAM, OLED Display, Buzzer, Edge Impulse Model	Identifies zebra crossings, stop signs, vehicles, and animals using camera input.	Visual display + buzzer alert
Optional Communication	Bluetooth / Wi-Fi (ESP32)	Enables data transfer or emergency alerts to smartphones.	Wireless connectivity

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